

Cloud Watch: A Monitoring tool



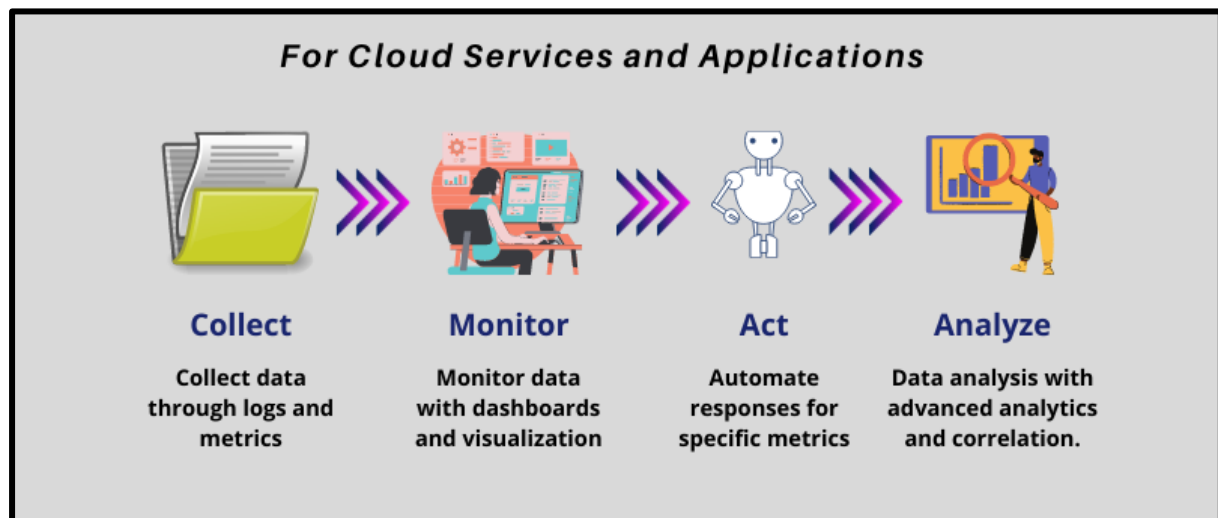
Amazon CloudWatch

“Amazon Cloud Watch is component of Amazon Web Services (AWS) that provides real time monitoring of AWS resources and customer applications”

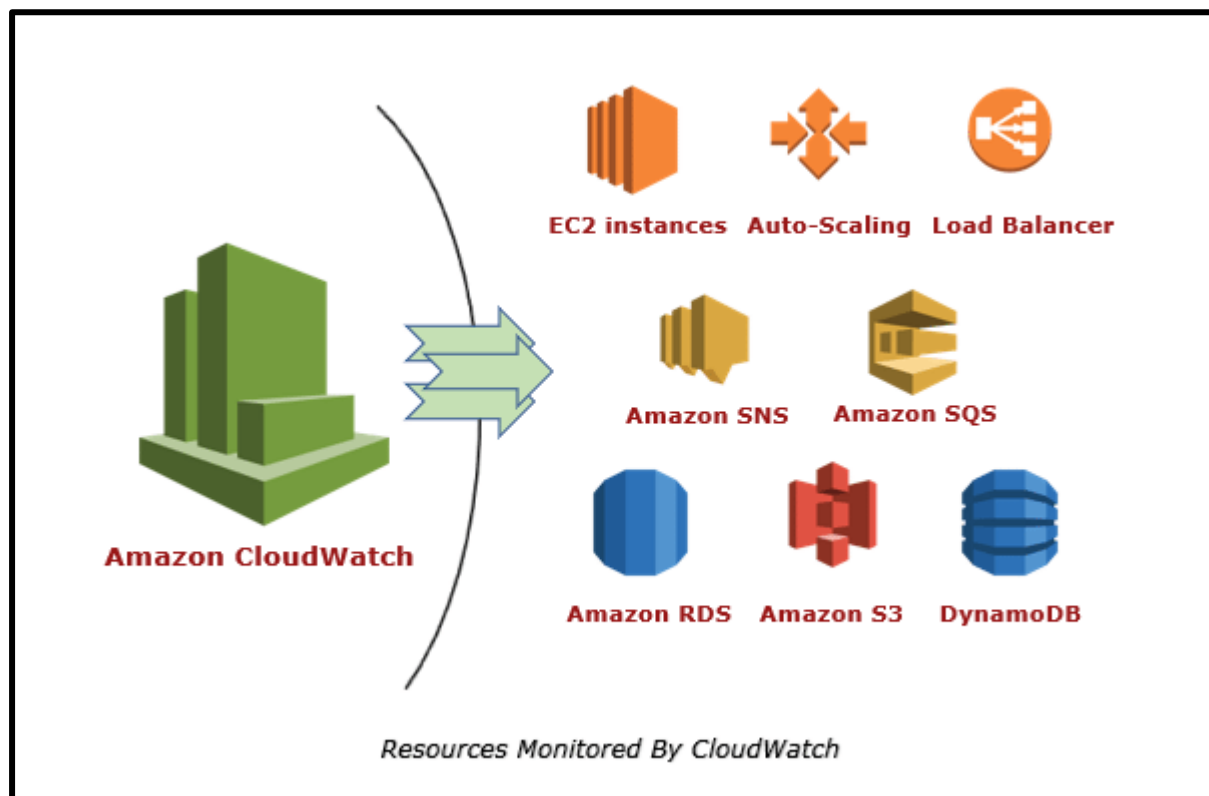
Types of Monitoring :---

Basic Monitoring	Detailed Monitoring
1. Free	1. Charged
2. Polls every 5min	2. Polls every 1min
3. Very Few Metrics	4. Wide Range of Metrics

What does Cloud Watch do ??



Resources Monitor by Cloud Watch :---



Cloud Watch **Concepts**:---



Metrics

A **metric** represents a time-ordered set of data points that are published to CloudWatch. Think of a metric as a variable to monitor, and the data points represent the values of that variable over time.



Dimensions

A **dimension** is a name/value pair that uniquely identifies a metric. They can be considered as categories of characteristics that describe a metric. We can assign up to 10 dimensions to a metric



Statistics

Statistics are metric data aggregations over specified periods of time. Aggregations are made using the namespace, metric name, dimensions within the time period you specify.



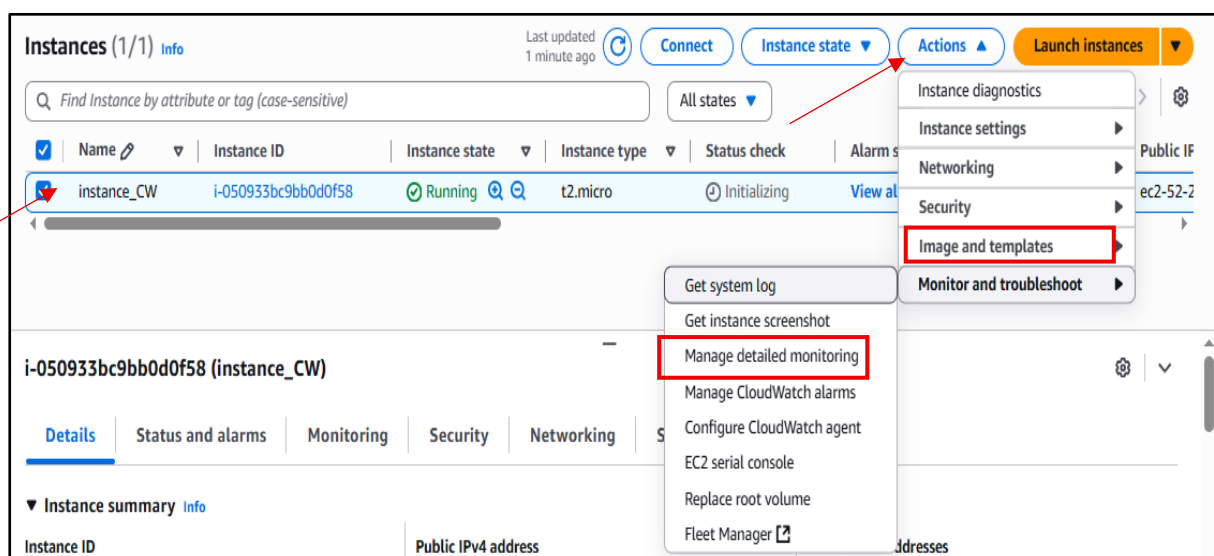
Alarm

An **alarm** can be used to automatically initiate actions on your behalf. It watches a single metric over a specified time period, and performs one or more specified actions.

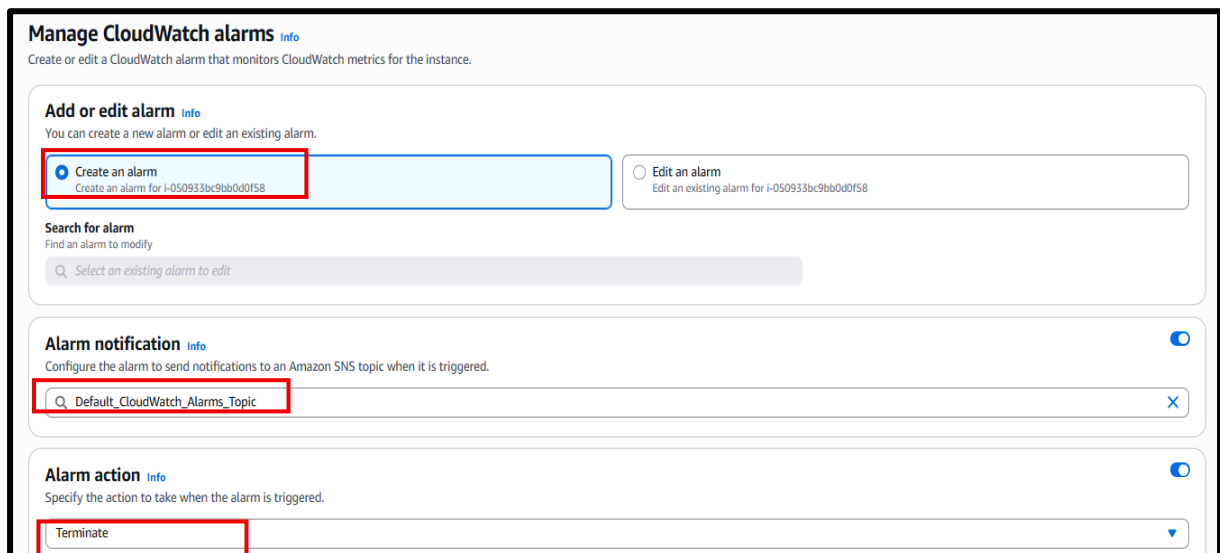


To create an alarm using the Amazon EC2 console

1. Open the Amazon EC2 console
at <https://console.aws.amazon.com/ec2/>.
2. In the navigation pane, choose **Instances** and create a instance.
3. Select the instance and choose **Actions, Monitor and troubleshoot, Manage CloudWatch alarms**.



4. On the **Manage CloudWatch alarms** detail page, under **Add or edit alarm**, select **Create an alarm**.
5. For **Alarm notification**, choose whether to configure Amazon Simple Notification Service (Amazon SNS) notifications.
6. For **Alarm action**, choose whether to specify an action to take when the alarm is triggered. Choose an action from the list.



The screenshot shows the 'Manage CloudWatch alarms' console. The 'Add or edit alarm' section has two radio buttons: 'Create an alarm' (selected) and 'Edit an alarm'. Below this is a search bar for existing alarms. The 'Alarm notification' section is enabled (toggle on) and shows a search bar with 'Default_CloudWatch_Alarms_Topic' entered. The 'Alarm action' section is also enabled (toggle on) and shows a dropdown menu with 'Terminate' selected.

7. For **Alarm thresholds**, select the metric and criteria for the alarm.
 - For example, to create an alarm that is triggered when CPU utilization reaches 80% for a 5 minute period, do the following:
 - a. Keep the default setting for **Group samples by (Average)** and **Type of data to sample (CPU utilization)**.

b. Choose **>=** for **Alarm when** and enter **0.80** for **Percent**.

c. Enter **1** for **Consecutive period** and select **5 minutes** for **Period**.

8. Choose **Create**.

Alarm thresholds

Specify the metric thresholds for the alarm.

Group samples by
Average

Type of data to sample
CPU utilization

Alarm when
>=

Percent
0.8

Consecutive period
1

Period
5 Minutes

Alarm name
awssec2-i-050933bc9bb0d0f58-GreaterThanOrEqualThreshold-CPUUtilization

Alarm description
Alarm on Instance i-050933bc9bb0d0f58: Triggered when CPUUtilization >= 0.8 for 1 consecutive 5-minute periods.

Sample metric data

Sample metric data for i-050933bc9bb0d0f58

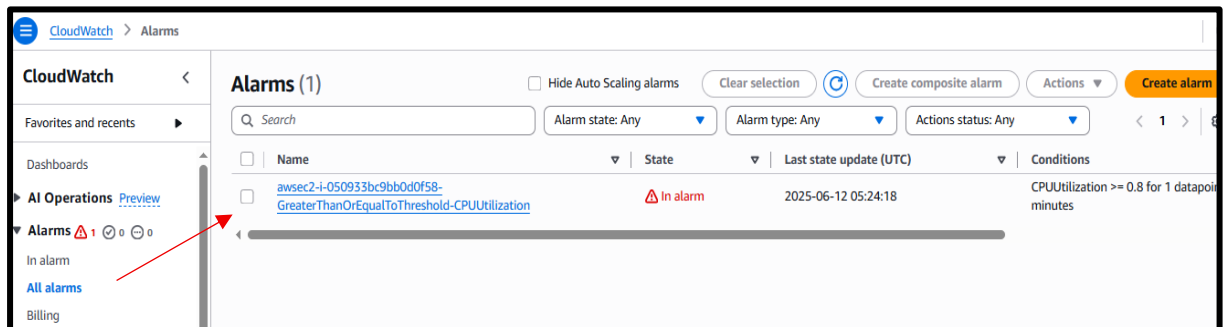
CPU utilization (Average)

0 0.8 3.97 7.94

02:30 02:45 03:00 03:15 03:30 03:45 04:00 04:15 04:30 04:45 05:00 05:15

[Cancel](#) [Create](#)

Alarm created ---



Watch after 5min instance got terminated :---

